

Survey Response Generation

**Generating Closed-Ended Survey Responses
In-Silico with Large Language Models**

Many *in-silico* surveys use closed-ended questions

Instructions

System Prompt, e.g.:

You are a political scientist predicting survey responses.

You only respond in the following

JSON format:

`{"response": <response_option>}`

Persona & Question

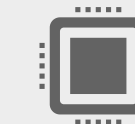
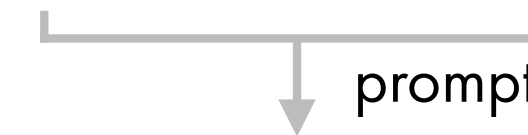
User Prompt, e.g.:

*Ideologically, I am a **liberal**.*

*I am **a woman**.*

*I am from **Kansas**.*

In the 2016 presidential election, I voted for



Large Language Model
e.g., OLMo 2 32B Instruct

Survey Response Generation Methods

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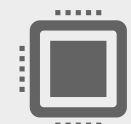
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*I am from **Kansas**.*

*In the 2016 presidential
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prompt



Large Language Model

e.g., OLMo 2 32B Instruct

Survey Response Generation Methods

**Token
Probability-
Based
Methods**

Match **token
probs** with re-
sponse options

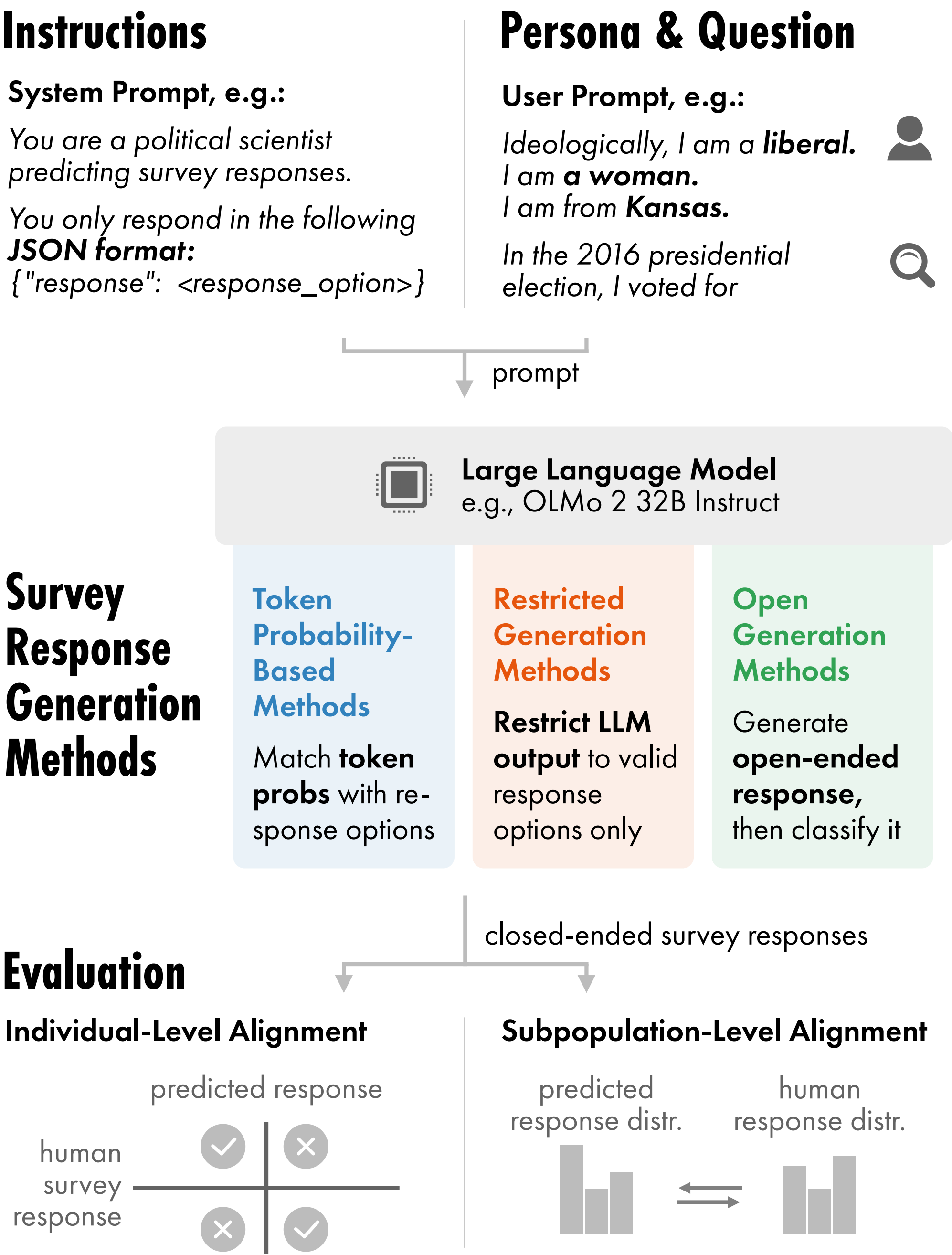
**Restricted
Generation
Methods**

**Restrict LLM
output** to valid
response
options only

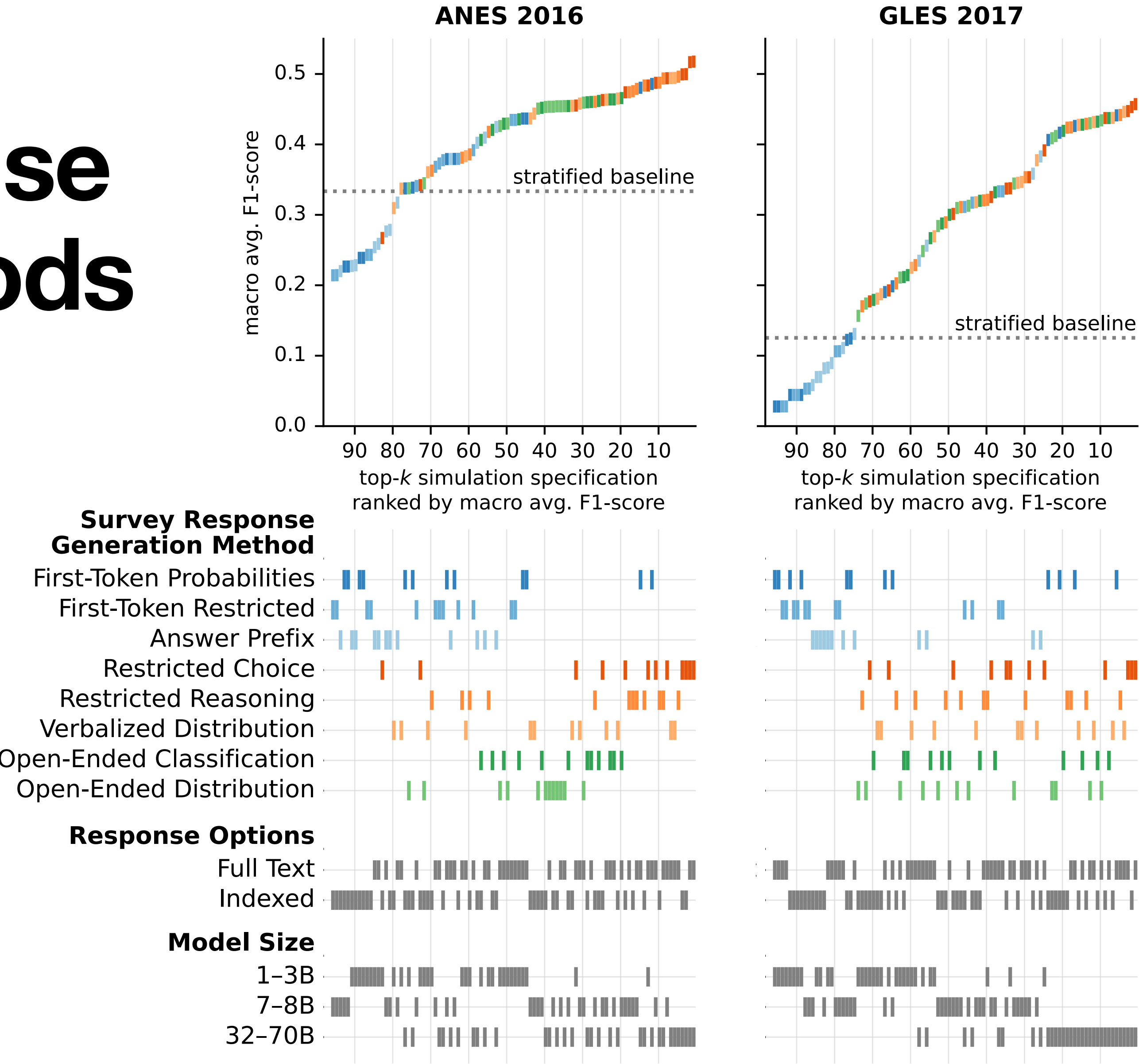
**Open
Generation
Methods**

Generate
**open-ended
response**,
then classify it

Individual- & Subpopulation-Level Evaluations



The Large Impact of Survey Response Generation Methods



Token Probability- Based Results Are Misaligned

OLS Regression
Coefficients

	Individual-Level		Subpop.-Level	
	Align-ment	Robu- stness	Align- ment	Global Align.
Intercept	-.397*	-.371*	.082	.007
First-Token Restrict.	.074	-.569*	-.316*	.342*
Answer Prefix	-.750*	-.260	.175	.147
Restricted Choice	<u>.763*</u>	.812*	-.360*	<u>.379*</u>
Restricted Reasoning	.996*	<u>.617*</u>	-.284*	.263*
Verbalized Distrib.	.756*	.447*	.183*	.384*

Open-Ended “Reasoning” Might Not Be Worth It

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Coefficients

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Restricted Reasoning	.996*	<u>.617*</u>	-.284*	.263*
Verbalized Distrib.	.756*	.447*	.183*	.384*
Open-Ended Classif.	.069	.403	-.174	.302*
Open-Ended Distrib.	.024	-.120	.008	.322*

Survey Response Generation

Generating Closed-Ended Survey Responses *In-Silico* with LLMs

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